

Guide to Hosting a Mesh Node for Property Owners, Building or Site Managers



Our volunteer-run, decentralized network relies on community hosts to build a resilient communication backup for emergency readiness and local community support.

Below are answers to common questions regarding the equipment, installation, and logistics.

The Basics:

What exactly is the node being used for?

It expands a free, open-source, community mesh network using compact, low-power radios that link to one another across the region. This infrastructure acts like a region-wide text messaging system, allowing people to communicate publicly or privately with anyone on the network. It enables reliable text-based communication in areas with limited cellular or internet coverage, during grid failures, natural disasters or major outages.

Who runs ColoradoMESH and where are the nodes deployed?

The ColoradoMESH community is a grassroots group of amateur radio operators, outdoor enthusiasts, hobbyists, and locals creating a resilient, decentralized communication system using affordable, off-the-shelf hardware. It is entirely community-run with many installations located on private homes, public works, commercial buildings, small businesses, and community partner locations across the Denver Metro area, Fort Collins, Colorado Springs, the Western Slope, and beyond.

Physical Impact and Installation:

What kind of equipment will be installed and how is it powered?

Installation typically consists of a small, weatherproof box containing a microcontroller and a battery, paired with a lightweight antenna and a small solar panel. The equipment is self-sufficient and operates autonomously for long periods, even during inclement weather. Units are compact and lightweight—usually weighing under 2 pounds with form factors ranging from the size of a standard red brick to a shoebox (excluding the vertical antenna).

What kind of antenna or cabling is involved?

Usually, a small antenna is attached directly to the node box itself. In other instances, a small external antenna may be mounted to a compact mast or pole. **Absolutely no cabling enters the building structure.** Typical non-invasive installation options include industrial magnets, U-bolts clamped around an existing structure/conduit, or attaching directly to a disused satellite dish mount. **No drilling or screwing** is performed unless explicitly discussed and agreed upon in advance.



Examples of Solar Powered Mesh Nodes

Will this use the building's power or increase the likelihood of lightning strikes?

No. One primary reason we deploy solar-powered nodes (even in fixed structure locations) is to completely isolate the hardware from the structure's electrical grid. Furthermore, because of its minimal size and footprint, it does not alter or increase the lightning risk profile of the property.

What is the expected maintenance requirement?

Extremely low. Nodes are explicitly engineered to run completely autonomously. Occasional brief site visits (e.g., once or twice a year) may be coordinated to perform firmware updates or standard hardware health checks. Access terms can be custom-tailored in writing to match site security requirements.

Signal and Technical Specifications:

What range and signal coverage will this node support?

Coverage depends heavily on structural elevation and surrounding terrain line-of-sight. Typically, a single node supports a range of 2 to 10 miles. For highly elevated or unobstructed vantage points, coverage can easily extend to 30+ miles, helping route critical text communications across long distances via automatic mesh relaying.

Will the equipment cause any interference with existing property systems?

Extremely unlikely. ColoradoMESH operates within the license-free Industrial, Scientific, and Medical (ISM) band (902–928 MHz). It operates on completely separate frequencies from standard building Wi-Fi (2.4 GHz / 5 GHz), cellular networks, or television systems, and functions at very low power levels. The hardware is fully FCC compliant and designed to be non-disruptive. Technical documentation or active testing can be provided upon request.

Liability, Terms and Logistics:

What are the liability implications?

Because the equipment is exceptionally lightweight (under 2 lbs.), entirely self-contained, and completely isolated from your building's electrical system, it poses no structural or utility risks to the property. The hardware remains the property and responsibility of the community volunteer network, and standard hold-harmless text can be coordinated if your legal or management team requires it.

Are there any permits, HOA, zoning, or regulatory issues to be aware of?

Permits are generally not required for small, non-commercial, temporary antennas. While HOA or corporate building rules occasionally restrict permanent exterior structural alterations, ColoradoMESH nodes are designed to be easily removable and use completely non-invasive magnetic or clamp mounts that require no drilling. We recommend obtaining written permission if you are a tenant or operating within a shared property space.

How long is the node expected to stay? Can it be removed at any time?

The node stays exactly as long as you want it to. We will work together transparently to determine a clear timeline or arrangement with no long-term obligations. You may request removal at any time; upon notification, we will coordinate a prompt removal window. **Removal leaves no permanent marks, residue, or structural damage.**

Community Partners and Support:

What are the benefits of hosting a community node?

Not only are you directly building an off-grid, decentralized communications network in your community, you can use it too! The combination of your location and structure height could serve as a critical link between nodes, and provide local mesh coverage.

Who else in Colorado hosts a community node?

Over 200 nodes have been deployed statewide by volunteers, with the goal of expanding the mesh coverage and enhancing resiliency. Check out the map of our growing network at <https://analyzer.meshcore.coloradomesh.org/#/map> or scan the QR code below.



**Thank you for considering hosting a
community node with ColoradoMESH!**

To learn more or coordinate a site evaluation, visit our
website www.coloradomesh.org



*No internet or cell towers needed—
just people-powered nodes keeping The Centennial State connected.*